2nd Order DEs: Substitutions

Questions 1–17

Question 1. It is given that $\dfrac{d^2y}{dx^2} + \dfrac{3}{x}\dfrac{dy}{dx} = x^2$.

a) Explain why we cannot use an auxiliary equation method to solve this differential equation.

b) Use the substitution $u = \dfrac{dy}{dx}$ to show that the equation becomes a 1st order differential equation in terms of u.

c) Use an integrating factor to show that $u = \dfrac{1}{6}x^3 + cx^{-3}$.

d) Hence find y in terms of x.

Question 2. It is given that $x\dfrac{d^2y}{dx^2} + \dfrac{1}{2}\dfrac{dy}{dx} = 3x^{3/2}$.

a) Explain why we cannot use an auxiliary equation method to solve this differential equation.

b) Use the substitution $u = \dfrac{dy}{dx}$ to show that the equation becomes a 1st order differential equation in terms of u.

c) Use an integrating factor to show that $u = \dfrac{3}{2}x^{3/2} + cx^{-1/2}$.

d) Hence find y in terms of x.

Question 3. It is given that $(2x+1)\dfrac{d^2y}{dx^2}+3\dfrac{dy}{dx}=4x+2$. Use the substitution $u=\dfrac{dy}{dx}$ to find the general solution to the differential equation.

Prerequisite knowledge for Q4: Reversing the Chain Rule, Calculus with Inverse Trigonometric Functions

Question 4. It is given that $(x^2+4)\dfrac{d^2y}{dx^2}+2x\dfrac{dy}{dx}=4x$, where $y=1$ and $\dfrac{dy}{dx}=0$ when $x=0$. Use the substitution $u=\dfrac{dy}{dx}$ to find a particular solution to the differential equation.

Prerequisite knowledge for Q5: Double Angle Formulae

Question 5. It is given that $\cos x \dfrac{d^2y}{dx^2} - 2\sin x \dfrac{dy}{dx} = \cos x$.

a) Find $\dfrac{d}{dx}\left(\dfrac{x}{2}\tan x\right)$.

b) It is given that $y = 2$ and $\dfrac{dy}{dx} = 3$ when $x = 0$. Use the substitution $u = \dfrac{dy}{dx}$ to find a particular solution to the differential equation.

Prerequisite knowledge for Q6: A-Level Maths Partial Fractions

Question 6. It is given that $(x+2)(x+3)\dfrac{d^2y}{dx^2} + \dfrac{dy}{dx} = (x+3)^2$, where $y = 1$ and $\dfrac{dy}{dx} = 3$ when $x = 0$. Use the substitution $u = \dfrac{dy}{dx}$ to find a particular solution to the differential equation.

Question 7.

a) Find $\frac{d}{dx}\left(x^4\frac{dy}{dx}\right)$.

b) It is given that $x^4(x^2+1)\frac{d^2y}{dx^2}+4x^3(x^2+1)\frac{dy}{dx}-2x^5\frac{dy}{dx}=2x(x^2+1)^2$. Use the substitution $u=x^4\frac{dy}{dx}$ to find the general solution to the differential equation.

Question 8. It is given that $(1+x^2)\frac{d^2y}{dx^2}-2x\frac{dy}{dx}=3-3x^2$.

a) Show that the substitution $u=\frac{dy}{dx}-3x$ transforms this differential equation into $(1+x^2)\frac{du}{dx}=2xu$.

b) Hence, given that $y=1$ and $\frac{dy}{dx}=2$ when $x=0$, find a particular solution to the differential equation.

Question 9. It is given that $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = e^{-3x}$.

a) Show that the substitution $u = \frac{dy}{dx} + 3y$ transforms this differential equation into $\frac{du}{dx} + 3u = e^{-3x}$.

b) By using an integrating factor, find the general solution of $\frac{du}{dx} + 3u = e^{-3x}$.

c) Hence find the general solution of $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = e^{-3x}$.

Transformation to Constant Coefficients

Question 10. It is given that $u = \ln x$. Show that

a) $\frac{dy}{dx} = \frac{1}{x}\frac{dy}{du}$

b) $\frac{d^2y}{dx^2} = \frac{1}{x^2}\left(\frac{d^2y}{du^2} - \frac{dy}{du}\right)$.

Question 11. It is given that $x^2\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - 4y = 0$, where $x > 0$. Use the substitution $x = e^u$ to find the general solution to the differential equation.

Question 12. It is given that $x^2\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - 8y = 4$, where $x > 0$. Use the substitution $x = e^u$ to find the general solution to the differential equation.

Question 13. It is given that $x^2\dfrac{d^2y}{dx^2}+3x\dfrac{dy}{dx}-3y=5x^2$, where $x>0$, and $y=2$ and $\dfrac{dy}{dx}=1$ when $x=1$. Use the substitution $x=e^u$ to find a particular solution to the differential equation.

Question 14. It is given that $x^2\dfrac{d^2y}{dx^2}-4x\dfrac{dy}{dx}+6y=12+10\cos(\ln x)$, where $x>0$.

a) Show that the substitution $x=e^t$ transforms this differential equation into $\dfrac{d^2y}{dt^2}-5\dfrac{dy}{dt}+6y=12+10\cos t$.

b) Hence find the general solution to the differential equation $x^2\dfrac{d^2y}{dx^2}-4x\dfrac{dy}{dx}+6y=12+10\cos(\ln x)$, where $x>0$.

Question 15. It is given that $x = e^{3t}$. Show that

a) $\dfrac{dy}{dx} = \dfrac{1}{3x}\dfrac{dy}{dt}$

b) $\dfrac{d^2y}{dx^2} = \dfrac{1}{9x^2}\left(\dfrac{d^2y}{dt^2} - 3\dfrac{dy}{dt}\right)$.

Question 16. It is given that $9x^3\dfrac{d^2y}{dx^2} + 2xy = x(\ln x)^2 + 20$, where $x > 0$.

a) Show that the substitution $x = e^{3t}$ transforms this differential equation into $\dfrac{d^2y}{dt^2} - 3\dfrac{dy}{dt} + 2y = 9t^2 + 20e^{-3t}$.

b) Hence find the general solution to the differential equation $9x^3\dfrac{d^2y}{dx^2} + 2xy = x(\ln x)^2 + 6$, where $x > 0$.

Question 17. It is given that $\cos^2 x\dfrac{d^2y}{dx^2} + 2\sin x\dfrac{dy}{dx} + 2y = \sin 2x\cos^3 x$, where $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.

a) Given that u is a function of x, show that the substitution $y = u\cos x$ transforms this differential equation into $\dfrac{d^2u}{dx^2} + u = \sin 2x$.

b) Hence find the general solution to the differential equation $\cos^2 x\dfrac{d^2y}{dx^2} + 2\sin x\dfrac{dy}{dx} + 2y = \sin 2x\cos^3 x$, where $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$.